KLE Technological University
Creating Value, Leveraging Knowledge
KLE TECH
Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

Minor-1 Project Report

on

Biometric Fingerprint Attendance System

By:

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled **" Biometric Fingerprint Attendance System"** is a bonafide work carried out by the student team of **"Ashwini M Dambal [02FE22BEC406], Pranjal Shindolkar [02FE22BEC416], Roopali Karadiguddi [02FE22BEC417], Snehal Sambaragi [02FE22BEC421]"**. The project report has been approved as it satisfies the requirements with respect to the mini project work prescribed by the university curriculum for B.E. (V Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2023-2024.

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ACKNOWLEDGMENT

I Acknowledged my guide In this beautiful moment, we take this opportunity to appreciate and express our gratitude's to all of them for their support and advice to us. Also to the well wisher who guid us for the successful completion of our project. In the course of completion our project we learnt the values of hard and perseverance, we have felt this experience as one of theose instance in life. Where in a person's character and the ability to work in a group reaches its sablime height. The success of any take is incomplete without the expression of sincere thanks and attitude to all the people, who have directly or indirectly contributed in the project thus, we would like to add few heartfelt words for the people who were a part of this numerous ways. Heartfelt thanks to Prof : SURESH MURGOD OF ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT and AMEY CHOUGULE for thier untiring and timely guidance in the completion of this project, and his active interest in our project helped us to achieve our goals. We offer our respectful and grateful attitude to Dr. S. F. Patil Principal of Dr. M. S. SHESHGIRI COLLAGE OF ENGINEERING AND TECHNOLOGY , BELAGAVI Campus for giving chance to share or ideas. We are thankful to our staff members for their valuable suggestions. Without guidance and supervision of all these people right from the very inception to the completion of the our project would have been incomplete.

-The project team B15

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ABSTRACT

Traditional attendance methods, such as paper registers, face numerous challenges including the risk of loss, damage, and the potential for proxy attendance (where one person marks attendance on behalf of another), compromising the reliability and security of attendance records. The IoT-based Biometric Fingerprint Attendance System offers a modern solution to these problems by utilizing fingerprint authentication, providing a secure method of recording attendance that resists fraud and unauthorized access, ensuring only the rightful individual can mark their presence and effectively eliminating buddy punching. Additionally, the system offers real-time attendance records, enhancing convenience and efficiency for administrators to monitor and manage attendance, thereby significantly improving the overall security, reliability, and efficiency of attendance management processes.

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Chapter 1

Introduction

Attendance management is a crucial aspect of organizational operations, impacting productivity, accountability, and security. Traditional methods, such as paper-based registers and manual logging, come with numerous challenges that can hinder their effectiveness and reliability. These methods are prone to physical loss, damage, and the possibility of proxy attendance, where one individual marks attendance for another. Such issues can lead to inaccurate attendance records and administrative inefficiencies, making it difficult to maintain the integrity of attendance data.
To address these challenges, advancements in technology, particularly the Internet of Things (IoT), have paved the way for more robust and secure solutions. The IoT-based Biometric Fingerprint Attendance System leverages biometric fingerprint authentication to provide a secure and precise means of tracking attendance. This system ensures that only authorized individuals can mark their attendance, effectively eliminating the possibility of proxy attendance and unauthorized access.
Moreover, the system offers real-time attendance records, allowing for timely monitoring and decision-making. This feature enhances the overall convenience and efficiency of attendance management, making it easier for administrators to oversee attendance and address any issues promptly. By addressing the limitations of traditional methods, the biometric fingerprint system represents a significant advancement in attendance management, offering enhanced security, reliability, and operational efficiency.

1.1 Motivation

1. Eliminate Proxy Attendance: Address the prevalent issue of buddy punching by ensuring that only the individual whose fingerprint is registered can mark their attendance.
2. Enhance Data Security: Provide a secure method of attendance recording that reduces the risk of unauthorized access and tampering.
3. Improve Accuracy: Mitigate the errors commonly associated with manual attendance logging, ensuring that attendance records are precise and reliable.
4. Streamline Administrative Processes: Reduce the administrative burden associated with maintaining and managing paper-based attendance systems.
5. Real-Time Monitoring: Enable real-time tracking and reporting of attendance, facilitating better oversight and quicker response to attendance-related issues.
6. Increase Efficiency: Automate attendance processes to save time for both employees and administrators, leading to increased operational efficiency.
7. Prevent Data Loss: Overcome the risk of loss or damage to physical attendance registers by digitizing attendance records.
8. Ensure Consistency: Maintain consistent and standardized attendance records across the organization, improving overall record-keeping and auditing processes.

1.2 Objectives

1. Enhance Security: Implement biometric fingerprint authentication to ensure that only authorized individuals can record their attendance, thereby preventing unauthorized access and buddy punching.
2. Improve Accuracy: Provide precise and reliable attendance records by eliminating human errors associated with traditional methods.
3. Ensure Real-Time Tracking: Enable real-time attendance tracking to facilitate timely monitoring and decision-making processes.
4. Increase Convenience: Simplify the attendance process for users and administrators through automated data collection and management.
5. Enhance Reliability: Develop a robust system that minimizes the risk of data loss or damage associated with paper-based methods.
6. Facilitate Integration: Ensure compatibility with existing organizational systems for seamless integration and enhanced functionality.

1.3 Literature survey

[1]Title-Automated Wireless Biometric Fingerprint System for Attendance Tracking
Objective: The objective of the system is to provide a secure and efficient means of attendance tracking using biometric technology, specifically fingerprint recognition. The system aims to eliminate issues such as buddy-punching and incorrect clocking by implementing a reliable biometric scheme.
Methods:
1.Fingerprint Enrollment Process:The system captures the fingerprint image and converts it into a character file stored in memory. Character files are combined to create a template for fingerprint matching. The template is sent back to buffer memory and then to the mobile app for storage and processing.
2.System Components: Utilizes Arduino ATmega 2560 with SPI/IIC protocols for interfacing. Features an OLED display for instructions and error messages. Relies on an Android phone to power the system and interact with the fingerprint module through Arduino for enrollment and attendance tracking.
3. Attendance Tracking Process: The system involves invoking the attendance function, searching data in flash memory, and returning a value upon a successful match. Data transmission between the Android app, Arduino, and various system components is facilitated for accurate attendance recording.
Limitations:
1. Biometric Scheme Concerns:
2. Users may have reservations about the Iris scanner's impact on eye health, leading to potential reluctance in adoption.
3. Password-based schemes have vulnerabilities such as shared passwords or system hacking, impacting user access.
4.System Complexity: The system involves multiple components like Arduino, fingerprint sensors, and mobile apps, which may increase complexity in setup and maintenance.
[2]Title-Portable Biometric Attendance System Using IoT.
Objective: Develop a smart device for automated attendance marking in educational institutions. Methods: Utilize hardware components and IoT technology for biometric identification and data transfer. Limitations: Requires a direct connection to a computer, lacks portability without external power supply, and may have limited data transfer range.
Methods:
1.Utilized hardware components like Arduino Nano, Fingerprint Scanner GT-511c3, NodeMCU.
2.Implemented software engineering principles for system development.
3.Integrated IoT technology for data transfer, storage, and display.
Limitations:
1.Requires a direct connection to a computer for data transfer.
2.Lack of portability without an external power supply.
3.Limited data transfer range with Zigbee module.
4.System may be bulky and time-consuming for student registration.
[3]Title-Automated Attendance System Using Biometrics and RFID Technology
Objective: The objective of this study is to implement an automated attendance system in universities by integrating biometric technologies such as face recognition and fingerprint scanning with RFID technology to improve accuracy and efficiency in attendance tracking.
1.Utilization of MATLAB, Arduino Software, and SDK program for fingerprint scanning.
2. Filling attendance tables with zeros to represent absence status.
3.Extracting features from images stored in the database directory.

4.Connecting web cameras and Arduino for data input.
Limitations:
1.Economical challenges in implementing the system.
2.Lack of encryption for database security.
3.Technical limitations such as using Arduino instead of Raspberry Pi for advanced systems.
4.Need for faster software programs than MATLAB for improved efficiency.
[4]Title-Enhanced Attendance Management System Based on Fingerprint
Objective: To develop a system that efficiently manages attendance using fingerprint recognition to prevent impersonation and reduce absenteeism.
Methods:
1. Utilized Waterfall methodology for linear development.
2. Implemented the system using Java programming language on NetBeans IDE.
3. Included phases like Requirement Analysis, System Design, Testing, Deployment, and Maintenance.
4. Incorporated tools like Java SDK, Griaule Biometrics SDK, and SecuGen SDK.
Limitations:
1.Existing system uses cumbersome paper work, prone to errors, and data loss.
2.Inaccuracy and frequent loss of data can lead to mismanagement of attendance policies.
3.Automated systems can reduce but not completely solve the problem of absenteeism and fraud.

Figure 1.1: Logo of KLE Technological University

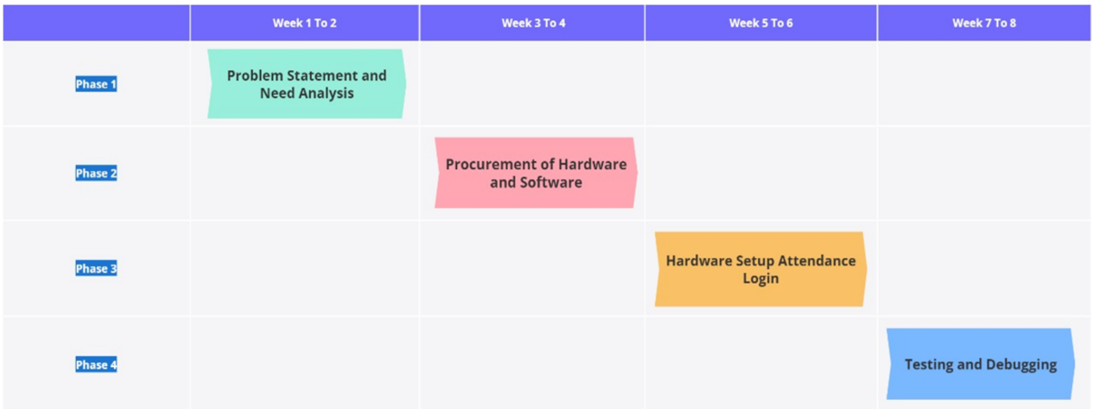
1.4 Problem statement

Design and implement a biometric attendance system using ESP8266 that integrates an OLED display and RS307 fingerprint sensor to provide real-time attendance tracking and management.

1.5 Application in Societal Context

1. Educational Institutions:
Schools, colleges, and universities can implement this system to ensure accurate attendance tracking, reduce truancy, and improve student accountability. This can lead to better educational outcomes and resource allocation.
2. Workplaces:
Corporations and small businesses alike can use the system to ensure punctuality and reliability among employees, enhancing productivity and reducing administrative burdens associated with traditional attendance management.
3. Healthcare Facilities:
Hospitals and clinics can track the attendance of medical staff accurately, ensuring that patients receive timely care and that staffing levels are adequately maintained.
4. Government Offices:
Public sector organizations can benefit from improved employee attendance tracking, which can enhance service delivery and public trust.
5. Event Management:
For conferences, workshops, and other events, this system can manage participant attendance effectively, ensuring security and streamlined check-ins.
6. Public Services:
Libraries, recreational centers, and other public services can use biometric attendance systems to manage staff and visitor attendance, improving service efficiency and security.vvvvvv

1.6 Project Planning and bill of materials



Sl No	Components	Price
1	NodMCU ESP8266	250
2	R305 Fingerprint Scanner Sensor Module	1000
3	0.96″ I2C OLED Disply	230
4	Bread board	50
5	Wires	20
6	USB cable	50
7	Total	1600

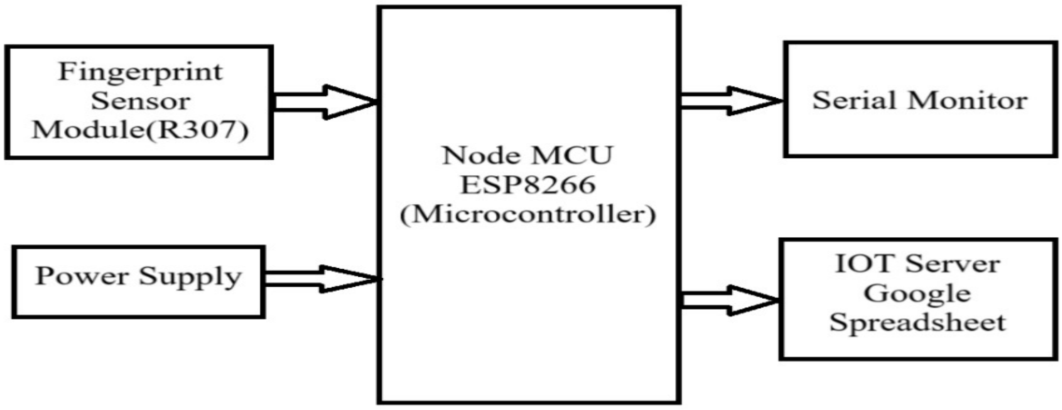
1.7 Organization of the report

1. Introduction
Overview of Biometric Systems: Importance and applications of fingerprint recognition.
Project Objectives: Goals of developing the Fingerprint Enrollment and Attendance Systems.
Scope: Hardware and software components, target applications, and expected outcomes.
Report Structure: Brief outline of the report contents.
2. Literature Review
Biometric Systems Overview: Principles and benefits of fingerprint recognition.
Comparison with Traditional Methods: Accuracy, security, and efficiency improvements.
Existing Solutions: Review of current systems, their features, and limitations.
3. Methodology
Requirements: List of hardware (Arduino, fingerprint sensor) and software (Arduino IDE, libraries).
System Architecture: Description and diagrams of system interactions.
Algorithms: Detailed explanation and flowcharts of the enrollment and attendance algorithms.
4. Implementation
Enrollment System:
Setup: Hardware and software initialization.
Process: Steps for capturing and storing fingerprints.
Error Handling: Strategies for managing errors during enrollment.
Attendance System:
Setup: Hardware and software configuration.
Tracking: Real-time attendance, relay control, and data transmission.
Error Handling: Error detection and resolution methods.
5. Results and Evaluation
Performance: Accuracy, response time, and reliability tests.
User Feedback: Usability and interface design feedback.
Objective Comparison: Analysis of how well objectives were met and challenges faced.
6. Discussion
Findings: Summary of strengths, weaknesses, and improvement areas.
Challenges: Technical difficulties and solutions.
Future Enhancements: Potential improvements and scalability options.
7. Conclusion
Summary: Achievements and significance of findings.
Lessons Learned: Insights and implications for future projects.
Final Thoughts: Overall impact of fingerprint-based systems.
8. References
Citations: List of all references used, following a specific citation style.
9. Report Timeline
Planning: Define objectives, review literature, finalize methodology.
Development: Implement systems, conduct testing, gather results.
Documentation: Write the report, prepare presentations, compile appendices.

Chapter 2

System design

2.1 Functional block diagram



2.2 Block diagram Explaination

Overview:
This fingerprint-based attendance system leverages a Node MCU ESP8266 microcontroller to authenticate users via a fingerprint sensor, then logs attendance data to an online server or Google Spreadsheet for easy access and management. The system ensures accurate and secure attendance tracking, with real-time monitoring capabilities.
Components and Interactions:

1. **Fingerprint Sensor Module (R307)**

   - Function: The R307 fingerprint sensor captures and processes the biometric data of a user's fingerprint. It converts the fingerprint image into a digital template that can be used for comparison and identification.

- Connection: This sensor is directly connected to the Node MCU ESP8266 microcontroller. The connection typically involves UART (Universal Asynchronous Receiver/Transmitter) communication, where the sensor sends the digital template data to the microcontroller.
- Operation: When a user places their finger on the sensor, it scans the fingerprint and sends the data to the Node MCU for processing.

2. Node MCU ESP8266 (Microcontroller)

- Function: The Node MCU ESP8266 is the central processing unit of the system. It is responsible for receiving the fingerprint data from the sensor, processing it to identify the user, and handling data communication with other components.
- Connections:

  Fingerprint Sensor Module: The microcontroller receives fingerprint data via UART communication.

  Serial Monitor: It sends data and status messages to the serial monitor for real-time debugging and monitoring. This connection is typically established through a USB interface when connected to a computer.

  IOT Server / Google Spreadsheet: Utilizes Wi-Fi capabilities to send attendance records to an online server or Google Spreadsheet. This is done through HTTP requests or API calls to update the database/spreadsheet.

  Power Supply: Receives power from the connected power supply, which ensures the microcontroller and connected peripherals are powered.
- Operation: Upon receiving fingerprint data, the microcontroller processes it to match against stored templates. If a match is found, it logs the attendance data (such as timestamp and user ID) and sends this information to the serial monitor and the IOT server or Google Spreadsheet.

3. Power Supply:

- Function: Provides the necessary electrical power to the Node MCU ESP8266 and the fingerprint sensor module.
- Connection: Directly supplies power to the Node MCU ESP8266, which in turn powers the connected fingerprint sensor.
- Operation: Ensures the system is continuously powered and operational.

4. Serial Monitor:

- Function: Acts as a real-time debugging and monitoring tool. It displays messages, data, and status updates sent from the Node MCU ESP8266.
- Connection: Receives data from the Node MCU via USB when connected to a computer.
- Operation: Used primarily during the development and troubleshooting phase to monitor the system's operation, check data being processed, and verify that communication with the fingerprint sensor and the IOT server is functioning correctly.

5. IOT Server / Google Spreadsheet

- Function: Serves as a cloud-based repository for attendance data. The Node MCU sends the recorded attendance (timestamp and user information) to this server or directly updates a Google Spreadsheet.
- Connection: The Node MCU uses its built-in Wi-Fi capabilities to connect to the internet and communicate with the IOT server or Google Spreadsheet.
- Operation: Once the attendance data is processed and ready, the Node MCU sends this data over the internet. This allows for remote storage and retrieval, making it easy to manage attendance records from anywhere with internet access.

Process Flow:

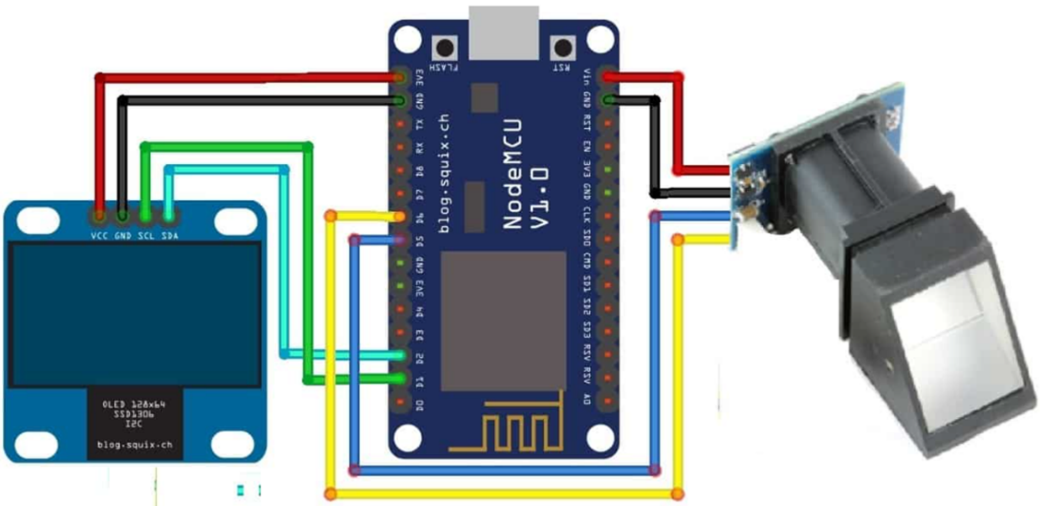
- Fingerprint Capture: A user places their finger on the Fingerprint Sensor Module (R307).
- Data Transmission: The sensor captures the fingerprint data and sends it to the Node MCU ESP8266.
- Processing: The Node MCU processes the fingerprint data to identify the user by comparing it against stored fingerprint templates.
- Logging Attendance: If a match is found, the Node MCU logs the attendance data.
- Real-time Monitoring: The attendance data is sent to the Serial Monitor for real-time feedback.
- Data Storage: The attendance data is also sent to the IOT Server / Google Spreadsheet for permanent storage.
- Power Management: The entire system is powered by the Power Supply, ensuring all components operate seamlessly.

2.3 Design alternatives

1. Choice of Fingerprint Sensor: Selecting a suitable fingerprint sensor is crucial to meet specific project requirements, such as higher resolution for accurate fingerprint recognition, faster response times, or onboard processing capabilities. Opting for a sensor with these features enhances accuracy, security through advanced algorithms, and ensures compatibility with future upgrades, thereby improving overall system reliability and performance.
2. Microcontroller Selection: Choosing Arduino Mega offers advantages in system architecture and scalability over Arduino Uno. With more digital and analog pins, Arduino Mega simplifies hardware integration for multiple sensors or peripherals without complex multiplexing. Its superior processing power supports efficient multitasking, accommodating tasks like sensor data acquisition, user interface management, and communication seamlessly.
3. Communication Protocol : Implementing MQTT protocol facilitates real-time data transmission in IoT applications. MQTT's lightweight nature ensures efficient communication between devices and cloud services, minimizing latency and optimizing bandwidth usage. Its reliability over unreliable networks ensures data integrity, crucial for continuous and secure data exchange, and supports scalability for expanding system capabilities.
4. User Interface Enhancement: Integrating a touchscreen display enhances user interaction and system functionality significantly. Unlike OLED displays, touchscreens provide intuitive navigation through interactive menus, enabling users to access records, perform tasks, and visualize data directly. This enhancement improves usability, offers features like graphical representation, and enhances operational efficiency across diverse environments.
5. Cloud Integration: Opting for AWS IoT Core or Azure IoT Hub enhances system capabilities and security compared to Google Sheets. These platforms offer scalable cloud storage, advanced analytics, and seamless integration with enterprise services, ensuring data security, facilitating comprehensive analysis, and supporting integration into existing IT infrastructures for enhanced efficiency and scalability.
6. Power Management: Implementing low-power design techniques optimizes energy efficiency, crucial for prolonged operation in resource-constrained environments. By minimizing power consumption during sensor operation and communication, the system extends battery life or reduces energy costs significantly. Effective power management ensures sustainable operation, ideal for remote deployments requiring continuous uptime.
7. Biometric Security Enhancements: Enhancing biometric security with multifactor authentication (e.g., fingerprint + PIN) strengthens user verification against threats like spoofing or unauthorized access. This approach increases security levels by requiring dual authentication factors, ensuring compliance with security standards and offering flexibility for diverse user preferences without compromising usability.

Chapter 3

Implementation details



1. Fingerprint-Based Attendance System

Initialization and Setup:

Libraries: Include necessary libraries such as Adafruit's fingerprint sensor library, WiFi library for network connectivity, and any required libraries for OLED display and HTTPS client. Variables and Constants: Define variables for controlling relays, storing WiFi credentials, Google Sheets API details, and sensor configurations. Serial Communication: Begin serial communication for debugging and monitoring purposes. WiFi Connection: Establish a WiFi connection to enable communication with Google Sheets or other cloud services. HTTPS Client: Initialize an HTTPS client for secure data transmission. OLED Display: Set up communication with an OLED display for real-time status updates. Fingerprint Sensor: Initialize communication with the fingerprint sensor, verify its presence, and retrieve parameters like capacity and security level. Main Loop:

Continuously monitor the fingerprint sensor for attendance purposes. Display a "Waiting..." message on the OLED screen while awaiting fingerprint input. Fingerprint Capture and Processing:

Capture fingerprint images and convert them into templates for matching against stored templates. Implement error handling for scenarios like no finger detected or communication errors with the sensor. Use algorithms to compare captured templates with stored ones and determine attendance status based on match results. User Authentication and Relay Control: Upon successful fingerprint match, toggle relays to indicate user entry or exit. Update real-time attendance records and transmit data securely to Google Sheets using HTTPS.

2. Fingerprint Enrollment System

Initialization and Setup:

Libraries: Include the Adafruit fingerprint sensor library and any necessary serial communication libraries. Variables and Constants: Define variables for fingerprint ID storage and sensor communication settings. Serial Communication: Begin serial communication for user interaction and debugging. Fingerprint Sensor: Initialize communication with the fingerprint sensor, set data rate, and verify sensor parameters. Main Loop:

Prompt the user to enter a unique fingerprint ID for enrollment. Initiate the enrollment process using the getFingerprintEnroll() function. Fingerprint Enrollment Process:

Guide the user through placing their finger on the sensor and capturing fingerprint images. Convert captured images into templates and create a combined fingerprint model for storage. Handle errors during image capture, template conversion, and storage to ensure successful enrollment. Helper Functions:

readnumber(): Read and validate user input for fingerprint ID entry. getFingerprintEnroll(): Implement the complete enrollment process including image capture, template conversion, model creation, and storage.

3.1 Specifications

NodeMCU ESP8266

- NodeMCU is predicated on the Esperessif ESP8266-12E Wi-Fi System-On-Chip. It is based on Lua-based firmware and is open-source.
- The ESP8266 is more updated and younger than Arduino, and therefore the ESP has stronger specifications than Arduino.
- Operating Voltage: 2.5 to 3.3V
- Operating current: 800 mA
- 3.3V 600mA on-board voltage regulation
- ESP8266 comes up with 2 switches one is reset and another one is flash button
- The board has build in LED indicator which is connected to D0 pin. The NodeMCU board also contains a CP2102 USB to UART module to convert the data from USB to serial so that it can be controlled and programmed via computer.
- The esp8266 has 4 power pins: One VIN pin for input power supply and three 3.3V pins for output power supply and also it has 3 GND pins.
- Esp8266 NodeMCU has 17 GPIO pins .
- ESP8266 NodeMCU has 2 UART interfaces, i.e. UART0 and UART1, which offer asynchronous communication, and may communicate at up to 4.5 Mbps. TXD0, RXD0, RST0 and CTS0 pins can be used for communication.
- ESP8266 has two SPI in slave and master modes. These SPIs also support the following general features: 4 timing modes of the SPI format transfer. Up to 64-byte FIFO buffer.
- Esp8266 has 4 channels of Pulse width modulation (PWM).

R305 Fingerprint Scanner Sensor Module

- This is a finger print sensor module with TTL UART interface for direct connections to microcontroller UART or to PC through MAX232 / USB-Serial adapter. The user can store the finger print data in the module and can configure it in 1:1 or 1: N mode for identifying the person.
- Fingerprint sensor type: Optical
- Sensor Life: 100 million times
- Static indicators: 15KVBacklight: bright green
- Interface: USB1.1/UART(TTL logical level)
- Image Capture Surface 15—18(mm)
- Verification Speed: 0.3 sec
- Scanning Speed: 0.5 sec
- Storage capacity: 250
- Voltage :3.6-6.0 VDC
- Working current: Typical 90 mA, Peak 150mA
- Matching Method: 1: N

0.96 I2C OLED Display

- This is a 0.96 inch blue OLED display module. The display module can be interfaced with any microcontroller using SPI/IIC protocols. It is having a resolution of 128×64. The package includes display board, display,4 pin male header pre-soldered to board.
- OLED (Organic Light-Emitting Diode) is a self light-emitting technology composed of a thin, multi-layered organic film placed between an anode and cathode. In contrast to LCD technology, OLED does not require a backlight. OLED possesses high application potential for virtually all types of displays and is regarded as the ultimate technology for the next generation of flat-panel displays.

3.2 Algorithm

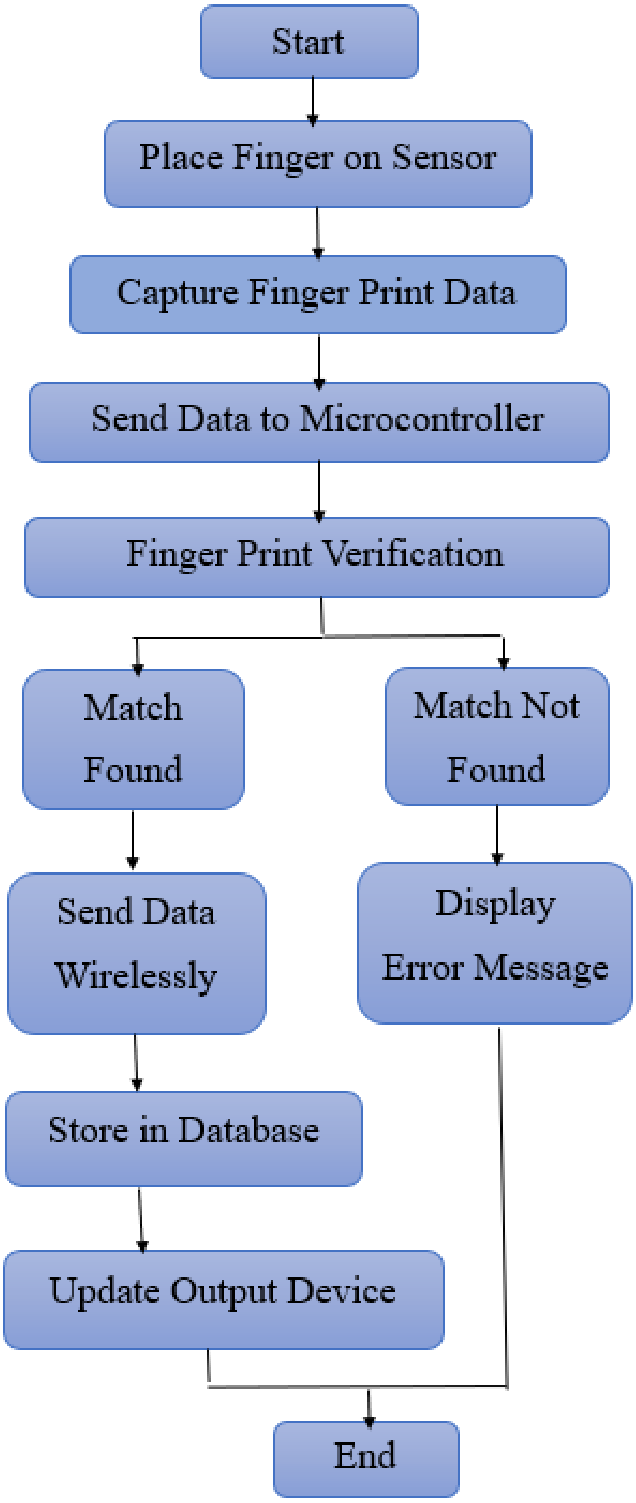
3.2.1 Algorithm of Main code:

1. Initialization and Setup
Define and Include Libraries:Include necessary libraries for fingerprint sensor, WiFi, OLED display, and HTTPS redirection.
Define Variables and Constants:Declare variables for relay control, user credentials, Google Script ID, and host details for Google Sheets.
Initialize Serial Communication:Begin serial communication for debugging and monitoring.
Connect to WiFi: Begin WiFi connection with provided SSID and password. Print connection status and IP address upon successful connection.
Initialize HTTPS Client:Create an HTTPS client object for secure communication with Google Sheets. Set security parameters and attempt to connect to the host.
Initialize OLED Display:Begin communication with the OLED display and clear the screen.
Initialize Fingerprint Sensor:Begin serial communication with the fingerprint sensor. Verify the presence of the fingerprint sensor and print its parameters.
Check Fingerprint Templates:Retrieve and print the number of fingerprint templates stored in the sensor.
2. Main Loop
Display Waiting Message:Clear the display and show a "Waiting..." message.
Capture Fingerprint:Call getFingerprintID() function to capture and process the fingerprint.
Display Success Message:If a valid fingerprint is detected, display a "Success" message.
3. Fingerprint Capture and Processing
Capture Fingerprint Image:Attempt to capture an image from the fingerprint sensor.Handle various cases like no finger detected, communication error, or successful image capture.
Convert Image to Template:Convert the captured image to a template for further processing. Handle various cases like messy image, communication error, or successful conversion.
Search for Fingerprint Match:Search for a matching fingerprint template stored in the sensor.Handle various cases like communication error, no match found, or successful match.
4. User Authentication and Relay Control
Match Found:If a matching fingerprint is found, determine the user ID and perform the following actions:
Control Relay:Toggle the relay state (HIGH/LOW) based on the user's previous state (entry/exit).
Update Google Sheets:Create a payload with user details and publish it to Google Sheets using the updatesheet function.
Indicate User Entry/Exit:Update the user's entry/exit status and send this information to Google Sheets.
No Match Found:If no matching fingerprint is found, display an error message and turn on the red LED.
5. Google Sheets Update Function
Initialize HTTPS Client:Create an HTTPS client object if it doesn't exist and ensure it is connected.
Create JSON Payload:Construct a JSON object with user details to send to Google Sheets.
Send Data to Google Sheets:Publish the JSON payload to Google Sheets and handle the response.
6. Helper Functions
getFingerprintID():Capture and process fingerprint images to search for a match. Control the LED and relay based on the match results and update Google Sheets.
getFingerprintIDez():Simplified version to quickly capture and process fingerprints.

3.2.2 Algorithm of student enrollement code:

1. Initialization and Setup
Include Libraries:Include the Adafruit Fingerprint library for fingerprint sensor operations.
Define Variables and Constants:Define a SoftwareSerial object for serial communication with the fingerprint sensor on non-Mega boards.
Define an Adafruit Fingerprint object to interface with the fingerprint sensor.
Declare a variable id to store the fingerprint ID.
Initialize Serial Communication:Begin serial communication at a baud rate of 115200 for debugging and monitoring.
Initialize Fingerprint Sensor:Set the data rate for the sensor serial port.
Verify the presence of the fingerprint sensor and print its parameters, such as status, system ID, capacity, security level, device address, packet length, and baud rate.
2. Main Loop
Prompt for Fingerprint ID:Prompt the user to enter the ID number (from 1 to 127) for the fingerprint to be saved. Read and validate the entered ID number.
Enroll Fingerprint:Call the getFingerprintEnroll() function to enroll the fingerprint for the specified ID.
3. Fingerprint Enrollment Process
Capture Fingerprint Image:Prompt the user to place a finger on the sensor. Attempt to capture an image from the fingerprint sensor. Handle various cases like no finger detected, communication error, imaging error, or successful image capture.
Convert Image to Template:Convert the captured image to a template for further processing. Handle various cases like messy image, communication error, fingerprint feature extraction failure, or successful conversion.
Prompt to Remove Finger:Prompt the user to remove the finger and wait for the sensor to detect no finger.
Capture Second Fingerprint Image:Prompt the user to place the same finger on the sensor again. Repeat the image capture and conversion process.
Create Fingerprint Model:Combine the two captured templates to create a fingerprint model. Handle various cases like communication error, fingerprint mismatch, or successful model creation.
Store Fingerprint Model:Store the created fingerprint model in the sensor's memory at the specified ID. Handle various cases like communication error, invalid storage location, flash memory error, or successful storage.
4. Helper Functions
readnumber():Read an integer number from the serial input, ensuring it is a valid entry.
getFingerprintEnroll():Implement the entire fingerprint enrollment process, including image capture, template conversion, model creation, and storage.

3.3 Flowchart



Chapter 4

Optimization

4.1 Introduction to optimization

Optimization in engineering and technology is the systematic process of refining systems, algorithms, or processes to enhance their efficiency, performance, and reliability. It involves analyzing the current state of a system, identifying areas for improvement, and implementing changes to achieve better outcomes. By focusing on reducing waste, minimizing resource consumption, and improving functionality, optimization aims to streamline operations and maximize output. This continual pursuit of efficiency not only enhances competitiveness and innovation but also ensures that technological solutions are sustainable, scalable, and resilient to meet evolving demands in today's dynamic landscape.

4.2 Types of Optimization

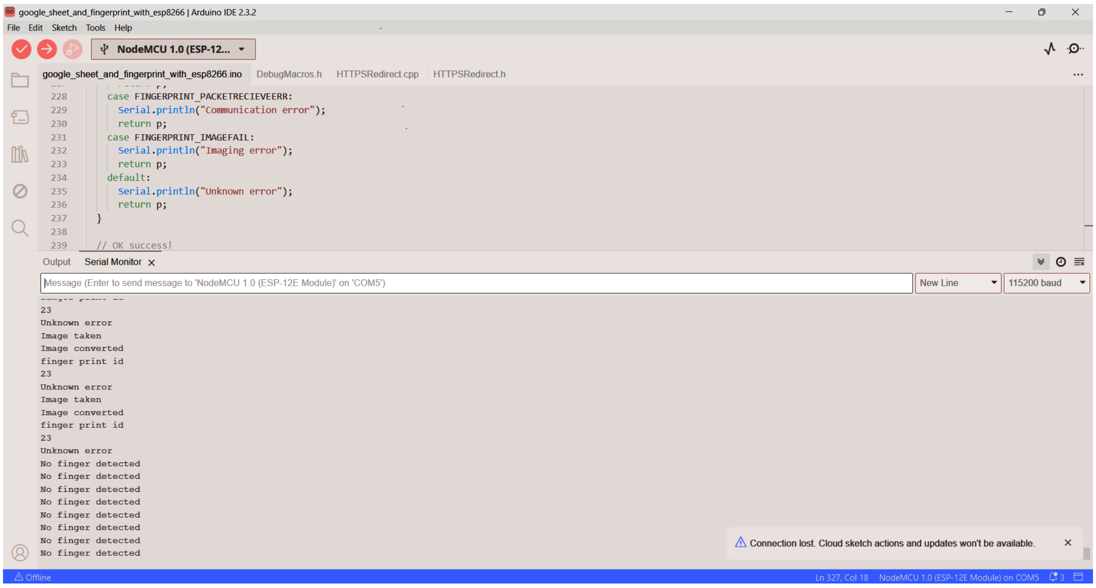
1 .Algorithmic Optimization:
Improving the efficiency of algorithms by refining their logic, reducing complexity, or implementing faster computational techniques.
2. Performance Optimization:
Enhancing the speed and responsiveness of software or hardware systems through code optimization, parallel processing, or hardware acceleration.
3. Resource Optimization:
Minimizing resource usage such as memory, storage, or bandwidth while maintaining or improving system performance.
4. Energy Optimization:
Designing systems to operate efficiently with minimal power consumption, crucial for mobile devices, IoT devices, and sustainable technologies.
5. Cost Optimization:
Streamlining processes to reduce operational costs, optimize supply chains, or improve manufacturing efficiency.

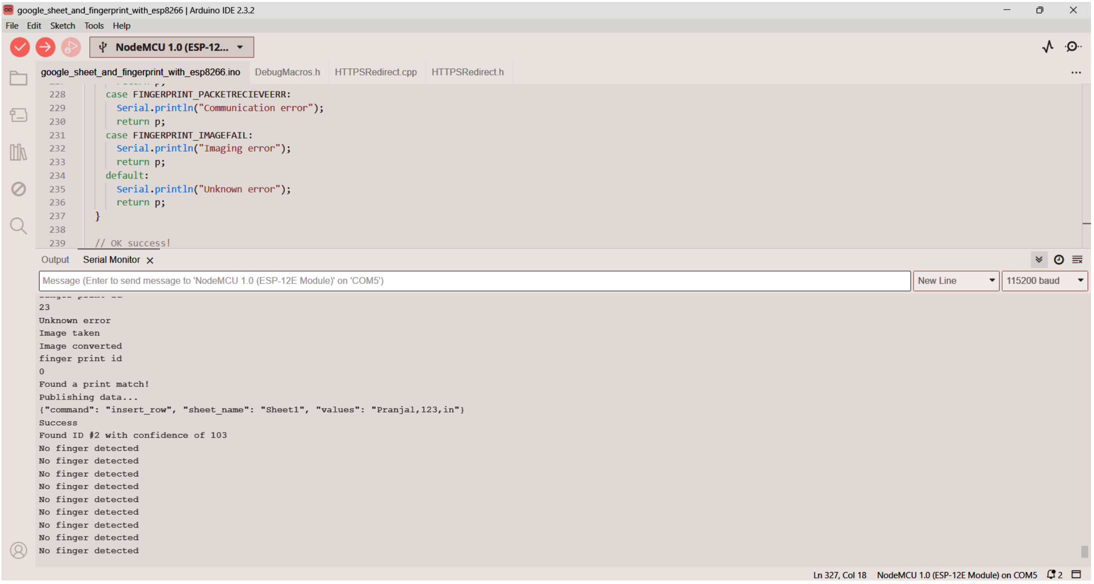
4.3 Optimization of the project

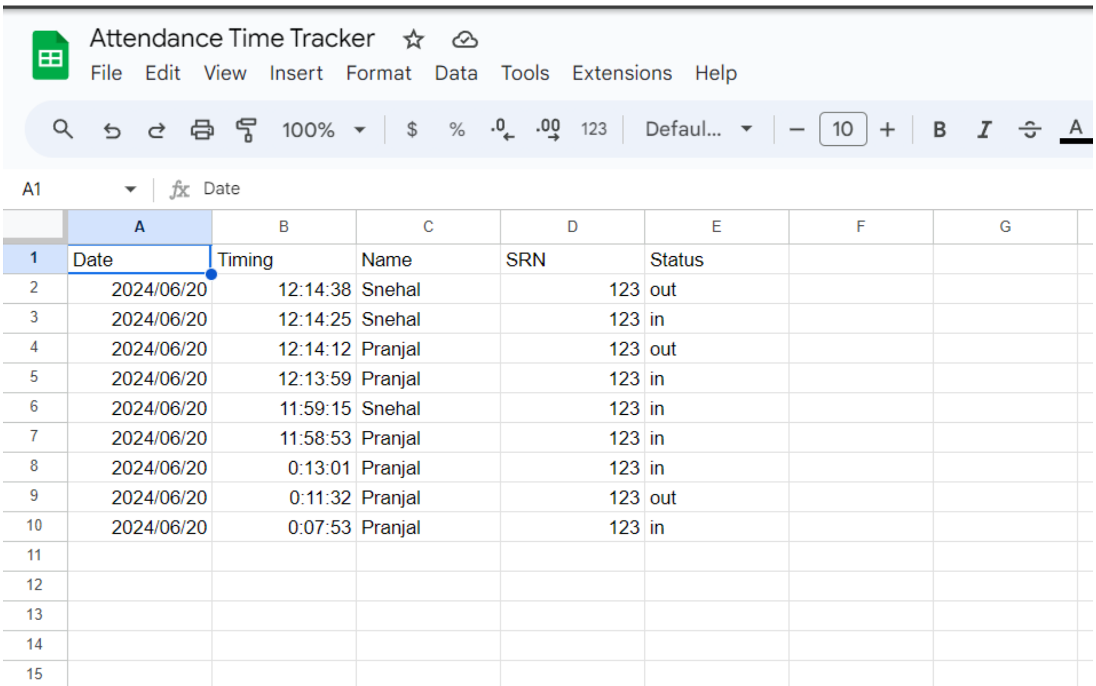
Efficient Code Execution:
Optimized Libraries: Utilize well-optimized libraries for fingerprint sensor operations, reducing latency and improving response times.
Efficient Serial Communication: Set an appropriate baud rate for serial communication, balancing speed and reliability to ensure quick data transfer between the Arduino and the fingerprint sensor.
Error Handling and User Feedback:
Comprehensive Error Handling: Implement detailed error handling to manage various scenarios such as communication failures, messy images, and template creation issues. This ensures the system can provide meaningful feedback and guide users through the enrollment process.
Clear User Instructions: Provide clear and concise instructions to users during the enrollment process, displayed via a connected OLED screen or serial monitor, to enhance user experience and reduce the likelihood of errors.
Memory Management:
Efficient Storage Utilization: Optimize the use of the fingerprint sensor's memory by ensuring that fingerprint templates are stored correctly and efficiently. This includes handling storage location errors and avoiding memory corruption.
System Scalability:
Support for Multiple Users: Ensure the system can handle a large number of users (up to the sensor's capacity), providing robust performance even as the number of enrolled fingerprints increases.
Modular Code Structure: Design the code in a modular fashion, allowing easy updates and expansions to the system, such as adding new features or integrating with other systems.
Power Management:
Low Power Consumption: Optimize the system for low power consumption, which is particularly important for battery-operated setups. This can involve putting the sensor and other components into low-power modes when not in use.
Security Enhancements:
Secure Data Transmission: Ensure that any data transmitted, especially sensitive fingerprint data, is encrypted and secure to prevent unauthorized access and tampering.
Authentication Protocols: Implement robust authentication protocols to verify the identity of the user before allowing access to stored fingerprint data or system settings.

Chapter 5

Results







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Chapter 6

Conclusions and future scope

6.1 Conclusion

The IoT-based Biometric Fingerprint Attendance System addresses the inherent limitations of traditional attendance methods by providing a secure, reliable, and efficient solution for tracking attendance. By utilizing fingerprint authentication, the system ensures that only authorized individuals can record their presence, effectively eliminating issues like proxy attendance and unauthorized access. The real-time attendance tracking capability enhances administrative efficiency, allowing for timely monitoring and decision-making. This system represents a significant advancement in attendance management, offering improved accuracy, security, and convenience for various applications across different sectors.

6.2 Future scope

1. Integration with AI: Incorporating artificial intelligence can enhance the system's ability to detect and adapt to usage patterns, further reducing the risk of fraud and improving accuracy.
2.Multi-Biometric Systems: Combining fingerprint authentication with other biometric methods, such as facial recognition or iris scanning, can provide even greater security and flexibility.
3.Cloud-Based Solutions: Implementing cloud-based data storage and processing can enhance the system's scalability, allowing for more efficient data management and accessibility across multiple locations.
4. Mobile Integration: Developing mobile applications that allow users to access their attendance records and receive notifications can improve user engagement and convenience.
5.Enhanced Data Analytics: Integrating advanced data analytics tools can provide deeper insights into attendance patterns, helping organizations make informed decisions about resource allocation and staffing.
6. Global Implementation: Expanding the system's use in diverse environments, including remote and underserved areas, can improve attendance tracking and accountability on a global scale.
7. Compliance and Privacy: Continuously improving the system to comply with evolving privacy regulations and standards will be crucial in maintaining user trust and protecting sensitive biometric data.

Chapter 7

References

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